

BRODY E. SNYDER

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EDUCATION

- **Purdue University, College of Science – John Martinson Honors College** *West Lafayette, IN*
Bachelor of Science, Computer Science – *Machine Intelligence Concentration* GPA: 3.82/4.0
Minors: Mathematics, Economics Exp. Graduation: May 2028
 - **Chick Evans Scholar:** Full tuition and housing scholarship awarded to golf caddies with academic excellence, leadership, and character
 - **CS Coursework:** Object-Oriented Programming, Data Structures & Algorithms, Systems Programming, Discrete Math

WORK EXPERIENCE

- **Software Engineering Intern – Iridium** *May 2026–Present*
 - Developing transceiver firmware for satellite communication, focusing on embedded hardware-software integration
 - Extending the 9704 developer kit with a solar charging subsystem, implementing battery-management and power-path firmware to improve energy autonomy in off-grid deployments and adding to CI/CD pipelines
- **AI Data Analytics Intern – Burbio** *July 2025–Sep. 2025*
 - Built AI-driven data extraction methods for K–12 school schedules using OCR, YOLO detection, and other computer vision techniques (SSIM, HOG, edge detection)
 - Streamlined school event data processing by automating extraction across 1,000+ schools, improving efficiency for partners

RESEARCH

- **3D Print Optimizing – CMSC** *Jan. 2025–Present*
 - Engineering a feedback control system linking computer vision, thermal sensing, and G-code execution to dynamically regulate an industry scale 3D printer's speed for optimized chemical bonds
- **Digital Twin – FLEX Lab** *Aug. 2024–Jan. 2025*
 - Contributed to a deep learning model for detecting cardiovascular failures using biosensor data
 - Developed a Flask-based web UI to visualize digital-twin outputs, enabling real-time inspection and analysis

PROJECTS

- **Numerai Prediction Model** *Aug. 2025–Present*
 - Building predictive models for financial time-series data as part of Purdue ML Club's participation in the Numerai competition
 - Collaborate with a team to optimize model generalization through feature neutralization, cross-validation, and hyperparameter tuning pipelines
- **CRUW Habit Tracking App** *May 2025–Aug. 2025*
 - Developed a full-stack habit tracker for individual and group accountability with a minimalistic design
 - Implemented a full-stack app with a React front end, JavaScript backend, and MongoDB Atlas database

LEADERSHIP

- **Vice President of New Scholars – Evans Scholars** *Jan. 2026–Present*
 - Schedule and lead study tables for 200+ hours to promote academic success
 - Organize community events fostering camaraderie and engagement
- **Event Lead & Test Writer – Science Olympiad** *Aug. 2024–Present*
 - Designed and wrote exams for Ecology and Wind Power events
 - Managed volunteers and coordinated logistics for 40+ schools and 600+ participants

SKILLS

- **Programming:** Java, Python, C, C++, HTML; experience with NumPy, Torchvision, PyTorch, Pandas
- **Tools:** Figma, Jira, Git, Fusion, Claude
- **Computer Vision:** OCR, HOG, SSIM, edge detection, YOLO integration